

Chapter 4: General Dynamic Games

ECON 5001

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Based on: *Game Theory* by Giacomo Bonanno

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Removing Perfect Information

Previously: every player knew the entire history at every node

That ruled out many important situations:

- Moves made **at the same time** as somebody else's
 - Duopoly competition (choosing price or quantity)
 - Repeated Prisoners' Dilemma
- Moves by “nature” (randomness)
 - Poker
 - Sealed bid auction w/out knowing others' values

Our goal:

1. Drop that assumption
2. Redefine “backwards induction” for these types of games

Two Columbus Businesses

Microbreweries often coordinate with food trucks.

Hypothetical story:

- **The Brewery** has one taproom and is thinking about opening a 2nd location
- **The Truck** wants to start working with this brewery

The Brewery decides first:

- **Stay**: No new location.
- **Expand**: Open a second taproom, either in **Franklinton** or in the **Short North**

The Truck sees the brewery's choice. Then both **simultaneously** pick:

- **Franklinton**
- **Short North**

Preferences

- Both want to land in the **same** neighborhood.
- The Brewery prefers **Franklinton**: cheaper rent, room for a patio
- The Truck prefers the **Short North**: late-night foot traffic on High Street

	Brewery	Truck
Brewery stays	2	0
Both in Franklinton	3	1
Both in the Short North	1	3
Different neighborhoods	0	0

Extra annual profit, in \$10,000s.

If the Brewery expands, this is an asymmetric coordination game

One Way To Model It

I could just write the associated strategic form:

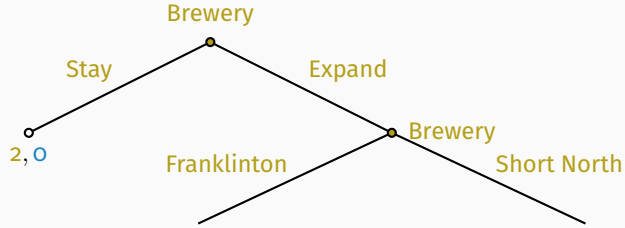
		Food Truck	
		Frkltn	ShrtNrth
Brewery	Stay Frkltn	2, 0	2, 0
	Stay ShrtNrth	2, 0	2, 0
	Expand Frkltn	3, 1	0, 0
	Expand ShrtNrth	0, 0	1, 3

...but that doesn't let me capture any backward-induction logic

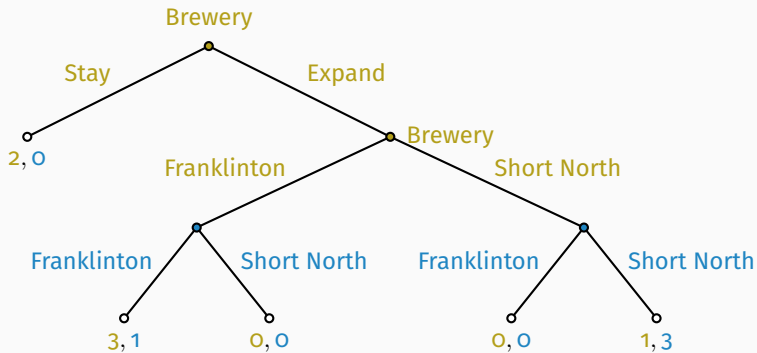
How To Draw It



How To Draw It

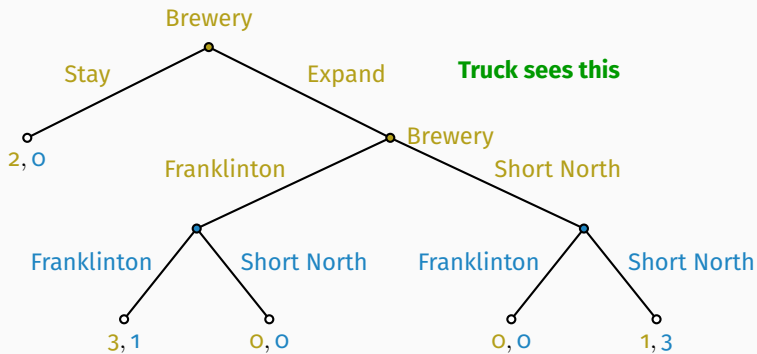


How To Draw It



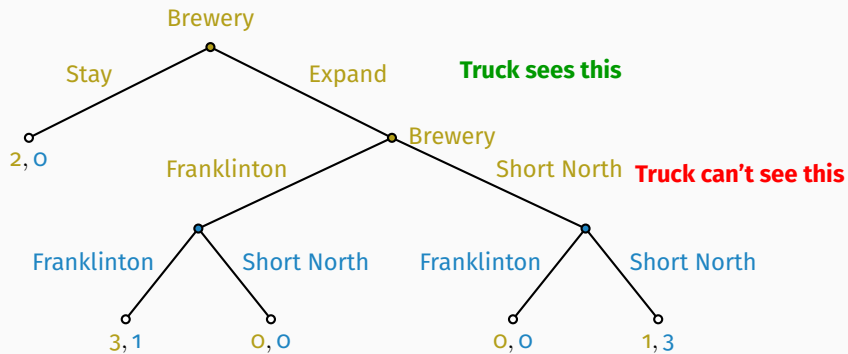
Problem: In this tree, The Truck **can** observe the Brewery's location choice.

How To Draw It



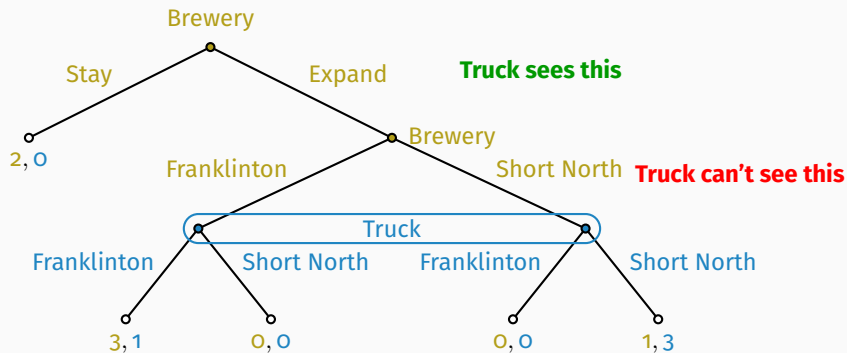
The first part is okay...

How To Draw It



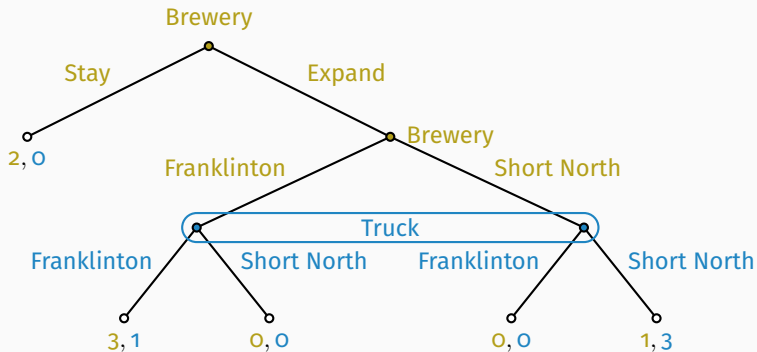
But the second part needs fixing...

How To Draw It



Draw an "Information Set": an oval indicating nodes that the Truck **cannot distinguish**.

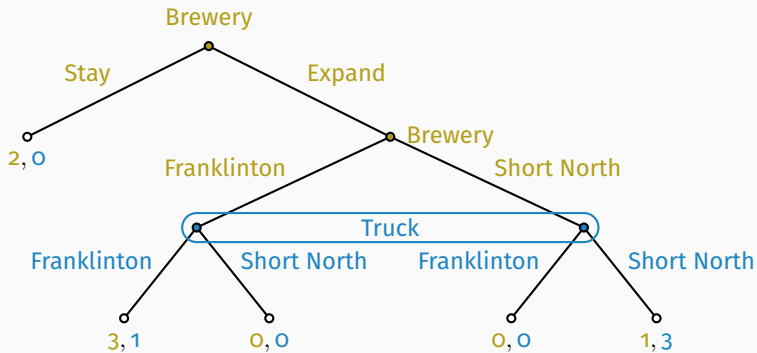
Let's Play It



Think about how you'd play...

Moblab time! Uses associated strategic form, but look at the tree here...

Let's Play It



Think about how you'd play...

The fact that the Brewery chose Expand tells us something...

Information Sets

Definition: Extensive-Form Game-Frame

Starts the same as a perfect-information game frame:

- a finite rooted directed tree
- players $I = \{1, \dots, n\}$, one assigned to every decision node
- actions on edges, no two edges out of a node sharing an action
- outcomes assigned to terminal nodes

The new piece: ★ For every player i , a **partition** $\mathcal{D}_i$ of her decision nodes D_i . Each cell of the partition is an **information set** (or “info set”).

Two nodes in the same info set: can't be distinguished

We write H for a single information set, so $H \in \mathcal{D}_i$.

Restriction: any two nodes in the same information set must offer *the same actions*.

Why must that hold?

A Note on Notation

Be careful with notation (Bonanno isn't):

- D_i is the set of **all** decision nodes belonging to player i
- $\mathcal{D}_i$ is the **partition** of D_i , a collection of sets
- $H \in \mathcal{D}_i$ is **one** information set, a single cell

Bonanno uses D for an info set, but D_i is also the set of all of i 's decision nodes. So let's use H instead.

Perfect Information Is the Special Case

Every information set is a singleton



Perfect information

Convention: No need to draw ovals around singleton infosets

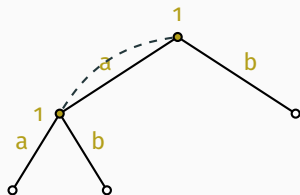
Some books/authors use dashed lines instead of ovals

Perfect Recall

We do assume “Perfect Recall”, which means two things:

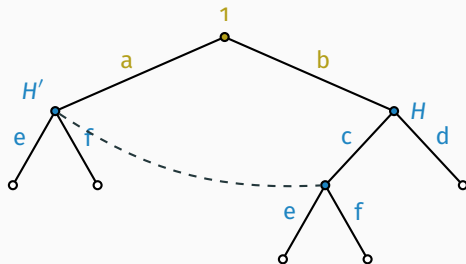
Condition 1

No node in an information set is a **predecessor** of another node in it.



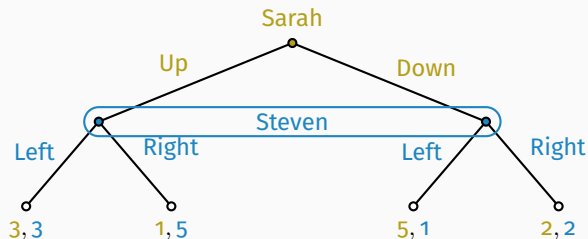
Condition 2

Players can't “forget” their past actions.
(See book for formal def'n)



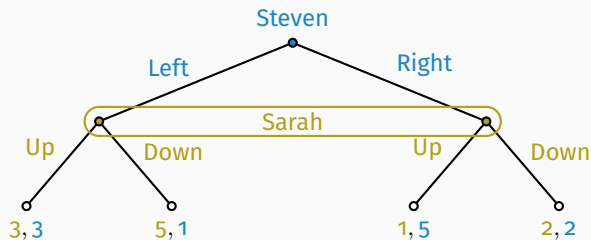
Simultaneous Moves

Simultaneous-move (strategic form) games can be drawn as extensive-form games:



Simultaneous Moves

Simultaneous-move (strategic form) games can be drawn as extensive-form games:



Simultaneity is not about time. It is about **what you know when you choose**.

What's a Strategy Now?

Strategy: Definition

Definition

A **strategy** for a player is still a **complete contingent plan**.

Now it specifies an action for every **information set** of the player.

Important: You must play the same action at all nodes of an info set

Tree to Table

We can still write any extensive-form game as its associated strategic-form game.

Just list the strategies as rows & columns

And the Nash equilibrium solution concept can still be used...
but it ignores backward induction-type logic

Next: an updated version of backward induction

		Food Truck	
		Frkltn	ShrtNrth
Brewery	Stay Frkltn	2, <u>0</u>	<u>2</u> , <u>0</u>
	Stay ShrtNrth	2, <u>0</u>	<u>2</u> , <u>0</u>
	Expand Frkltn	<u>3</u> , <u>1</u>	0, 0
	Expand ShrtNrth	0, 0	1, <u>3</u>

But First... Subgames

IKEA and Wayfair, Again

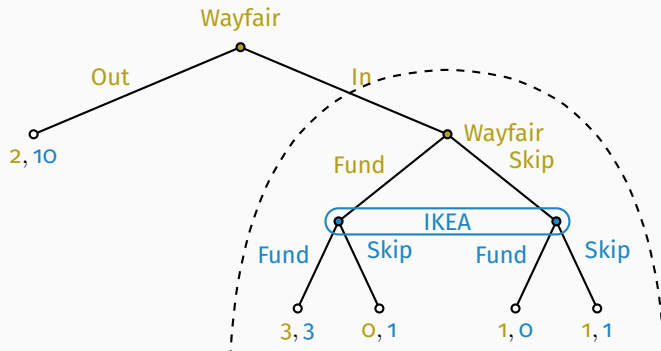
IKEA is the established furniture store at Polaris, earning \$10 million a year in profit. Wayfair decides whether to open across I-71.

If Wayfair opens, the two stores **simultaneously** decide whether to fund a joint campaign advertising Polaris as a furniture destination.

- Both fund it: shoppers drive up from all over. Both gain.
- Neither funds it: nothing changes.
- One funds it alone: too small to work, and the money is gone.

That second stage is a coordination game too. Unlike the Brewery and the Truck, both stores rank its two equilibria the same way.

The Whole Game and a Subgame



Annual profit in \$millions. First number Wayfair, second IKEA. The circle marks the only proper subgame.

Subgames

A **subgame** is a piece of the tree that could stand alone as a game.

Informal procedure:

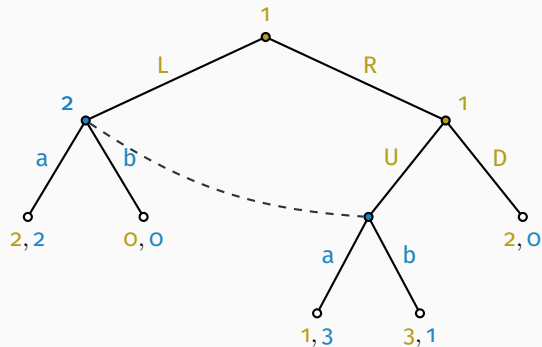
1. Pick a decision node x whose information set is a singleton (e.g. $\{x\}$)
2. Draw a big circle around x and everything that follows it

One rule: your circle can't "cut through" any information sets

Every game is a subgame of itself.

Anything smaller is called a **proper** subgame.

Example



Can we draw a subgame starting after R ?

Formal Definition: Subgame

Let $S(x)$ be the set containing x and all its successors.

$S(x)$ is a **subgame starting at x** if:

1. the information set containing x is the singleton $\{x\}$, and
2. for every information set H in the game, if $H \cap S(x) \neq \emptyset$ then $H \subset S(x)$

It is a **proper** subgame if x is not the root.

A subgame is **minimal** if it contains no other subgame strictly inside it.

Subgame-Perfect (Nash) Equilibrium

Definition

Let s be a strategy profile for the whole game G , and G' a subgame. Write $s|_{G'}$ for the **restriction** of s to G' (what actions to take inside of G')

Definition

s is a **subgame-perfect equilibrium** of game G if, for **every** subgame G' of G , the restriction $s|_{G'}$ is a Nash equilibrium of subgame G' .

SPNE

The whole game is one of its own subgames, so $SPNE(G) \subseteq NE(G)$ (just like *BI*)

SPNE is a **refinement** of *NE*

The Algorithm

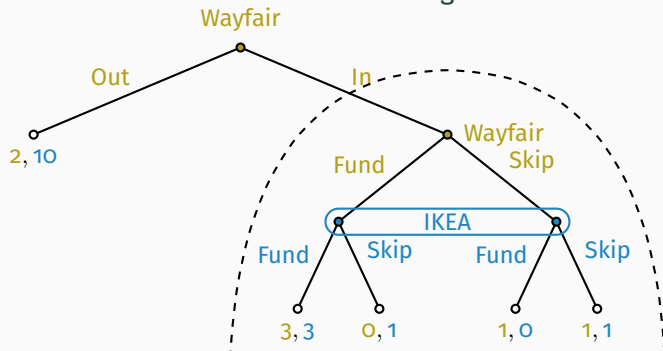
1. Find a **minimal** subgame.
2. Solve it. Find its Nash equilibria and pick one.
3. Replace the subgame with its NE payoff vector. You now have a smaller game.
4. Repeat until no proper subgames remain.

If any subgame had several Nash equilibria, run the procedure again for each NE of the subgame

If some subgame has *no* Nash equilibrium, then the game has no *SPNE*.

Back to The Example

Find the SPNE of this game:



Step 1: Solve the Subgame

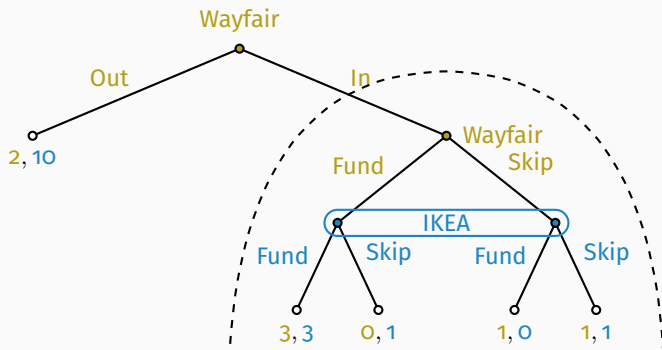
		IKEA	
		Fund	Skip
Wayfair	Fund	<u>3</u> , <u>3</u>	0, 1
	Skip	1, 0	<u>1</u> , <u>1</u>

Two Nash equilibria: (Fund, Fund) with payoffs (3, 3), and (Skip, Skip) with payoffs (1, 1).

So we have to run the algorithm **twice**.

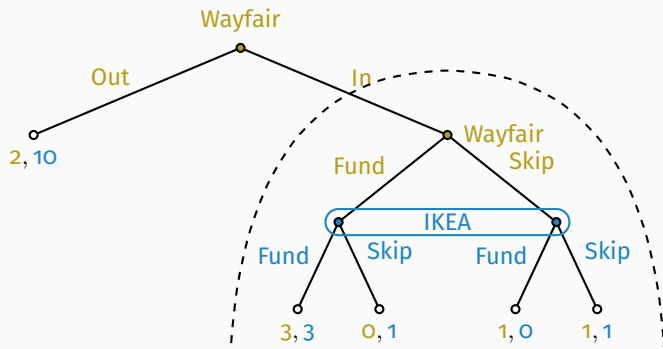
Step 2: Replace and Solve

Using the **(Fund, Fund)** equilibrium.



Step 2: Replace and Solve

Using the **(Skip, Skip)** equilibrium.



Two Subgame-Perfect Equilibria

$((\text{In}, \text{Fund}), \text{Fund})$ and $((\text{Out}, \text{Skip}), \text{Skip})$

In the second SPNE Wayfair never opens, *because* it expects no campaign.

- Both firms would prefer the first equilibrium
 - It Pareto dominates the second SPNE
- Subgame perfection does **not** pick between them

The Strategic Form

The associated strategic form:

		IKEA	
		Fund	Skip
Wayfair	(Out Fund)	2, 10	2, 10
	(Out Skip)	2, 10	2, 10
	(In Fund)	3, 3	0, 1
	(In Skip)	1, 0	1, 1

The Strategic Form

The associated strategic form:

		IKEA	
		Fund	Skip
Wayfair	(Out Fund)	2, <u>10</u>	<u>2</u> , <u>10</u>
	(Out Skip)	2, <u>10</u>	<u>2</u> , <u>10</u>
	(In Fund)	<u>3</u> , <u>3</u>	0, 1
	(In Skip)	1, 0	1, <u>1</u>

Three Nash equilibria.

The strategic form has **three** Nash equilibria.

NE Profile	Subgame play	Subgame perfect NE?
((In, Fund), Fund)	(Fund, Fund)	yes
((Out, Skip), Skip)	(Skip, Skip)	yes
((Out, Fund), Skip)	(Fund, Skip)	no

The third NE: If Wayfair plays Out, then Fund and Skip don't matter, so both can be part of a NE

SPNE: *if the subgame were reached* (eg, Wayfair trembles) then Wayfair would Skip.

A Bigger Example

Bonanno Figure 4.9. Find the SPNE.

10/8

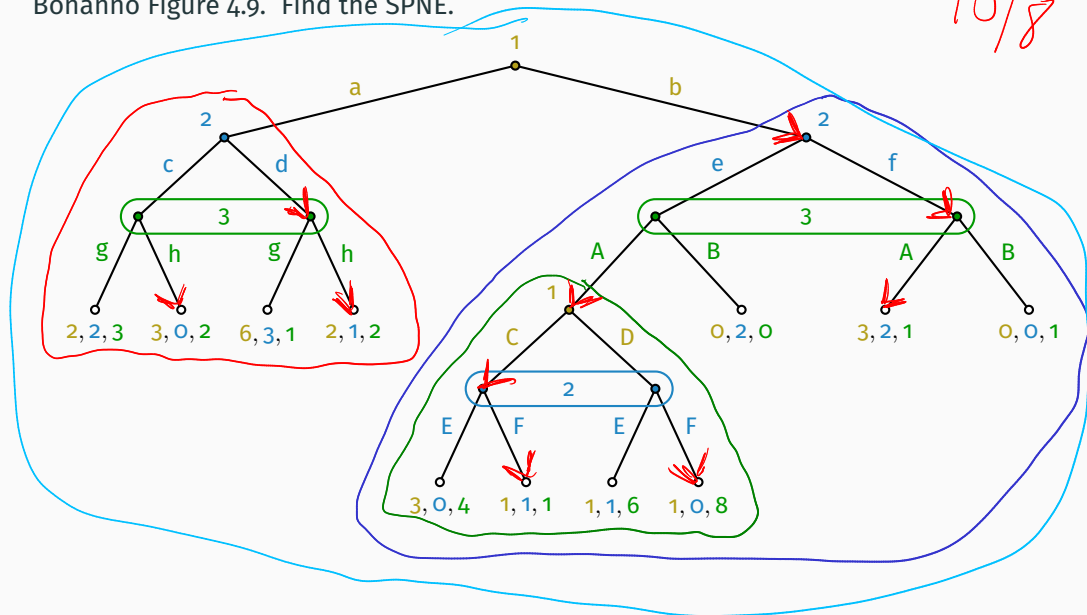
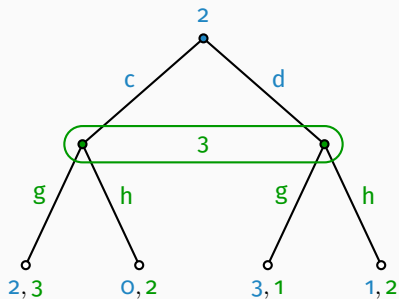


Figure 4.10: A Minimal Subgame



		Player 3	
		g	h
Player 2	c	2, <u>3</u>	0, 2
	d	<u>3</u> , 1	<u>1</u> , <u>2</u>

Only P2 and P3 move here, so only their payoffs are shown.

NE of this subgame: $(\cdot, d, h)$

Figure 4.11: Replace the Subgame

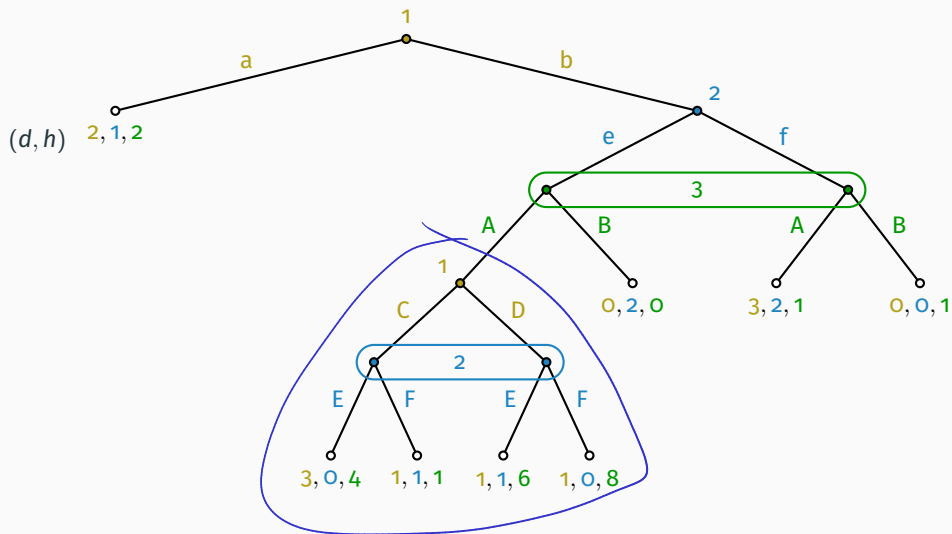
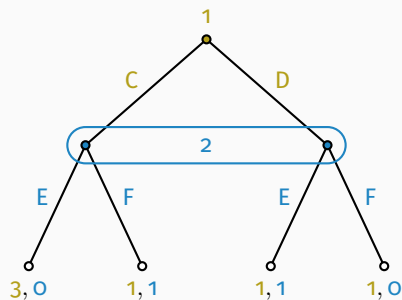


Figure 4.12: The Next Minimal Subgame



		Player 2	
		E	F
Player 1	C	<u>3</u> , 0	<u>1</u> , <u>1</u>
	D	1, <u>1</u>	<u>1</u> , 0

NE of this subgame: (C, F, ·) *payoffs (1, 1)*

Figure 4.13: Replace the Subgame

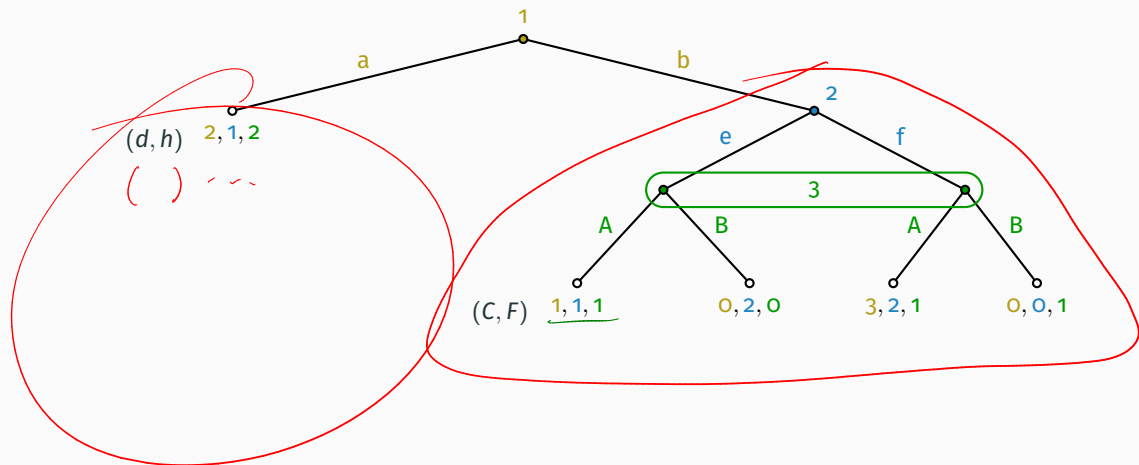
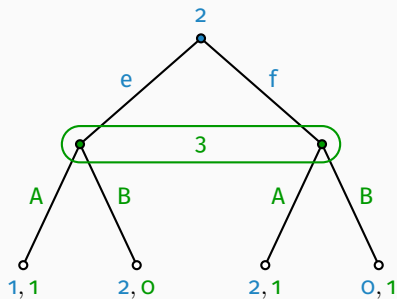


Figure 4.14: The Last Proper Subgame

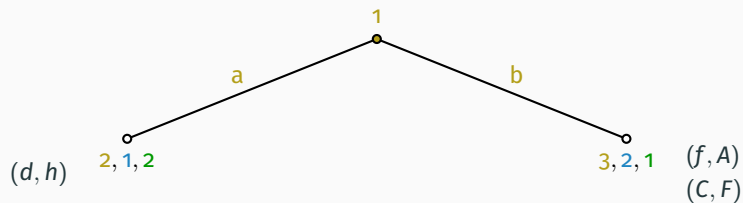


		Player 3	
		A	B
Player 2	e	1, <u>1</u>	<u>2</u> , 0
	f	<u>2</u> , <u>1</u>	0, <u>1</u>

NE of this subgame: $(\cdot, f, A)$

Bonanno's table labels the row player "Player 1". That's a typo: it's Player 2.

Figure 4.15: What's Left

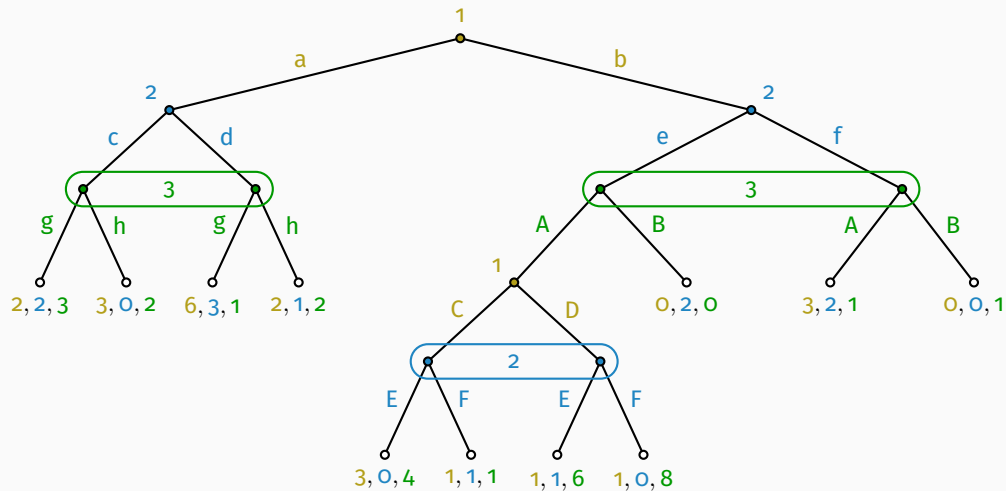


Player 1 chooses: **b**

SPNE: $\left(\underbrace{\left(\underline{\quad \text{b} \quad}, \underline{\quad \text{C} \quad} \right)}_{P1}, \underbrace{\left(\underline{\quad \text{d} \quad}, \underline{\quad \text{f} \quad}, \underline{\quad \text{F} \quad} \right)}_{P2}, \underbrace{\left(\underline{\quad \text{h} \quad}, \underline{\quad \text{A} \quad} \right)}_{P3} \right)$

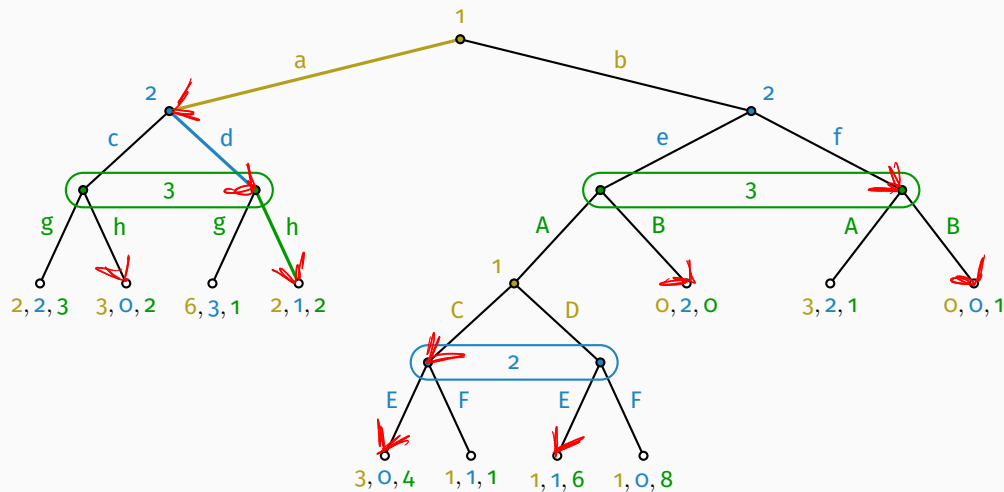
The SPNE in the Whole Game

Draw the SPNE in the full tree:



A NE That Is Not SPNE

$((a, C), (d, f, E), (h, B))$ is a NE. Why isn't it a SPNE?



Good practice question: write the associated strategic form and solve for its NE ($4 \times 8 \times 4$: four tables!)

Try This One Yourself

Bonanno's **Figure 4.16** is another three-player game with several subgames.

We won't do it in class. Work through it on your own:

1. Find the proper subgames. Which are minimal?
2. Solve each minimal subgame and replace it with its NE payoffs
3. Write the strategic form of what's left and find its NE
4. Patch the pieces together into full strategy profiles

Hint: this one has more than one SPNE.

Bonanno solves it right after the figure (Figure 4.17), so you can check your answer.

Three Things Worth Remembering

- In a **perfect-information** game, $SPNE = BI$
- If a game has **no proper subgames**, every Nash equilibrium is subgame perfect.
- $SPNE \subseteq NE$

Chance Moves (aka Moves By Nature)

When Nobody Chose It

Sometimes what happens isn't anyone's decision. A card is dealt, a part fails, a customer walks in.

We model this with a fictitious player: **Nature** (or **Chance**).

- Nature moves at some nodes
- Each of Nature's actions carries a **probability**
- Nature has no payoffs and no objectives, so it isn't really a player

Everything else (information sets, strategies, subgames) is unchanged.

Three Cards

Three cards: one black, two red. Shuffled, face down.



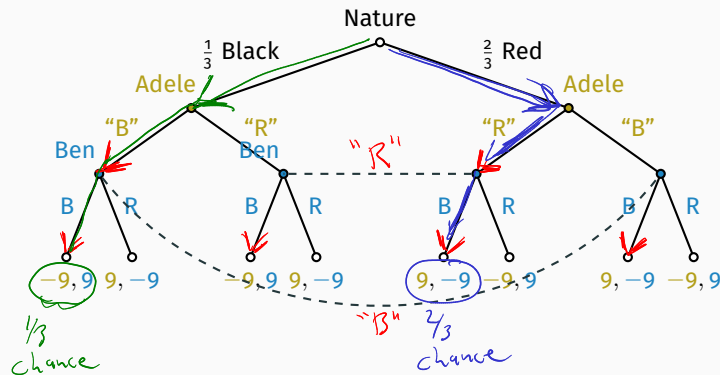
Nature moves first, turning up the top card: black with probability $\frac{1}{3}$, red with probability $\frac{2}{3}$.

Then the players move:

- Adele looks at the top card without showing Ben
- She tells Ben “black” or “red”, truthfully or not
- Ben then guesses the true color
- Correct guess: Adele pays Ben \$9. Wrong: Ben pays Adele \$9.

Ben has two information sets, one for each thing Adele might say, and in each he is unsure what the card actually is.

The Card Game



Nature moves first. Adele's label is what she says; Ben's is what he *guesses*. First payoff Adele, second Ben.

Ben has two information sets, one for each thing Adele might say. In each, he knows what he heard but not what the card is.

A New Difficulty

Take the profile: Adele always tells the truth, Ben always guesses black.

What's the outcome? It depends on the card, so the outcome is itself a gamble:

Outcome	Adele pays \$9	Ben pays \$9
Probability	$\frac{1}{3}$	$\frac{2}{3}$

outcome of
strategic profile
is
a lottery

We call this a **lottery**. To build a strategic form we need to know how players rank *lotteries*, not just outcomes.

Chapter 5 answers that properly. For now, one shortcut.

Expected Value

Definition

The **expected value** of the money lottery $\begin{pmatrix} \$x_1 & \$x_2 & \cdots & \$x_n \\ p_1 & p_2 & \cdots & p_n \end{pmatrix}$ is

"average" payoff



$$x_1p_1 + x_2p_2 + \cdots + x_np_n$$

prizes/outcomes on top
probability of each

$$\begin{pmatrix} \$5 & \$15 & \$25 \\ \frac{1}{5} & \frac{2}{5} & \frac{2}{5} \end{pmatrix}$$

has expected value

$$\frac{5(\frac{1}{5}) + 15(\frac{2}{5}) + 25(\frac{2}{5})}{1 + 6 + 10} = 17$$

Adele's lottery

$$\begin{pmatrix} -\$9 & \$9 \\ \frac{1}{3} & \frac{2}{3} \end{pmatrix}$$

has expected value

$$\frac{(-9)(\frac{1}{3}) + (9)(\frac{2}{3})}{-3 + 6} = +3$$

Risk Neutrality

Definition

A player is **risk neutral** if she considers a money lottery to be just as good as its expected value.

So a risk-neutral player ranks lotteries by their expected value, and we can use that number as her payoff.

$$\begin{array}{ccc} u(L_1) = 17 & u(L_2) = 16 & u(L_3) = 16 \\ L_1 = \begin{pmatrix} \$5 & \$15 & \$25 \\ \frac{1}{5} & \frac{2}{5} & \frac{2}{5} \end{pmatrix} & L_2 = \begin{pmatrix} \$16 \\ 1 \end{pmatrix} & L_3 = \begin{pmatrix} \$0 & \$32 & \$48 \\ \frac{5}{8} & \frac{1}{8} & \frac{1}{4} \end{pmatrix} \\ & & 0\left(\frac{5}{8}\right) + 32\left(\frac{1}{8}\right) + 48\left(\frac{1}{4}\right) \\ & & 0 + 4 + 12 = 16 \end{array}$$

A risk-neutral player ranks these: $L_1 > L_2 \sim L_3$

This is a convenience, not a claim about people. Risk neutrality lets us finish Chapter 4 without the machinery of Chapter 5.

Why the Assumption Is Doing Work

L_2 and L_3 have the same expected value, \$16.

- L_2 pays \$16 for certain
- L_3 pays nothing five times out of eight, and sometimes \$48

A risk-neutral player is **indifferent** between them.

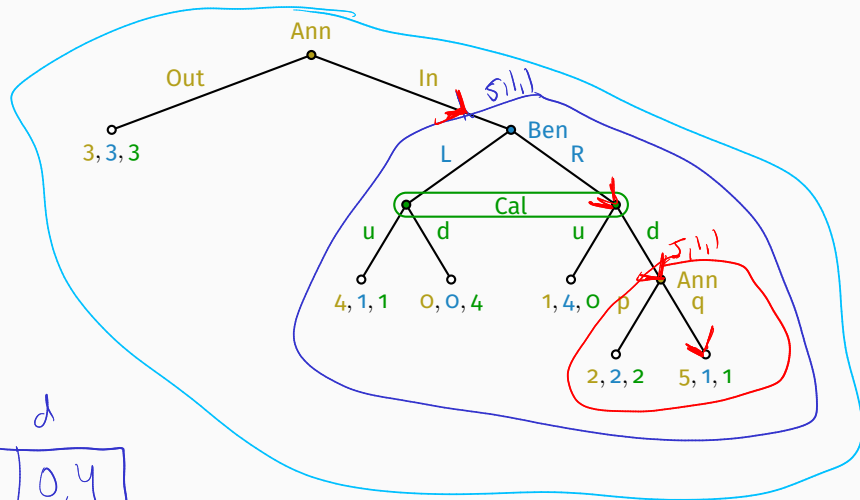


Most people are not. Ask yourself which you'd take, and notice that your answer is information the expected value throws away.

Recovering it is exactly what Chapter 5 is for.

Let's Work One Out

A Three-Player Dynamic Game



	u	d
L	1, 1	0, 4
R	4, 0	1, 1

SPE: $(In, q), R, d$ payoff: $(5, 1, 1)$

A Three-Player Dynamic Game

Its strategic form. Ann's strategy is a pair: the root, then her later node.

		Ben				Ben	
		L	R			L	R
Ann	(Out p)	3, <u>3</u> , <u>3</u>	<u>3</u> , <u>3</u> , <u>3</u>	(Out p)	<u>3</u> , <u>3</u> , <u>3</u>	3, <u>3</u> , <u>3</u>	
	(Out q)	3, <u>3</u> , <u>3</u>	<u>3</u> , <u>3</u> , <u>3</u>	(Out q)	<u>3</u> , <u>3</u> , <u>3</u>	3, <u>3</u> , <u>3</u>	
	(In p)	<u>4</u> , 1, 1	1, <u>4</u> , 0	(In p)	0, 0, <u>4</u>	2, <u>2</u> , <u>2</u>	
	(In q)	<u>4</u> , 1, 1	1, <u>4</u> , 0	(In q)	0, 0, <u>4</u>	<u>5</u> , <u>1</u> , <u>1</u>	← SPNE
		Cal: u				Cal: d	

A Three-Player Dynamic Game

1. List every subgame. Why is there none starting at Cal's left node?
2. How many strategies does each player have?
3. Run the subgame-perfect equilibrium algorithm. Solve the smallest subgame first.
4. Find a Nash equilibrium of the whole game that is **not** subgame perfect.

What Have We Learned, Charlie Brown?

- An **information set** collects nodes a player cannot tell apart; perfect information is the case where all are singletons
- A **strategy** gives one choice per information set, not per node
- A **subgame** starts at a singleton information set and cuts no information set
- **Subgame-perfect equilibrium**: Nash in every subgame. Solve minimal subgames first, replace, repeat
- **Nature** handles chance; risk neutrality ranks lotteries by mean

↑
expected value

Next: what if players aren't risk neutral?

- Bonanno, *Game Theory*, Chapter 4
- Exercises: §4.6.1 (imperfect information), §4.6.2 (strategies), §4.6.3 (subgames), §4.6.4 (subgame-perfect equilibrium), §4.6.5 (chance moves)
- Exercise 4.2 is the entry game with a simultaneous second stage, close to the IKEA example, with more to it
- Solutions in §4.7